Lecture 14: Synthesis
Solar System in Context, Astrobiology

Introduction to Planetary Science

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Tidally locked rocky planet atmospheric circulation. Wordsworth & Kreidberg (2022)

Learning Objectives

By the end of this lecture, you will be able to:

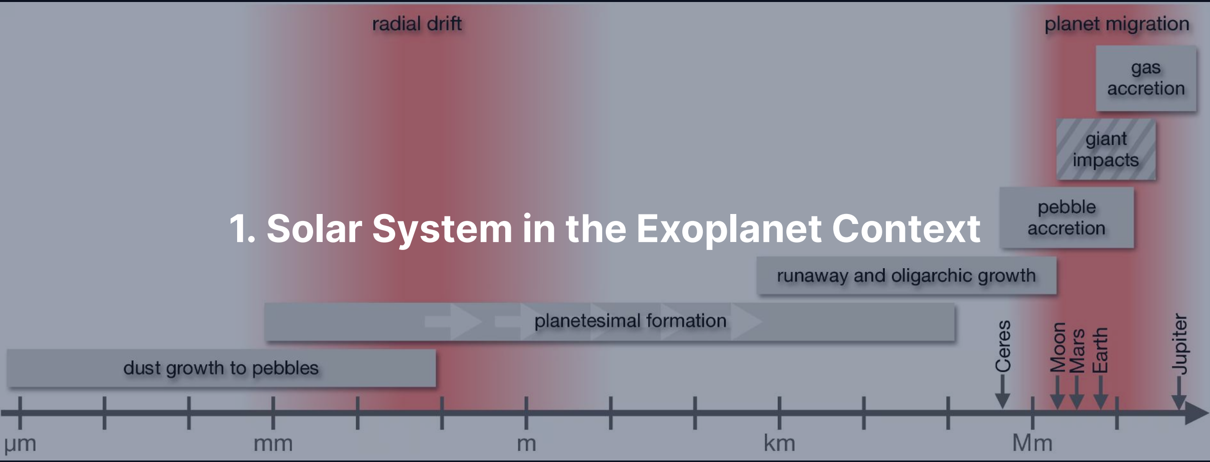
- Place our solar system within the observed exoplanet population.
- Evaluate habitability as a coupled *systems* property, not a single condition.
- Derive the classical habitable-zone boundaries from stellar luminosity and the liquid-water condition.
- Identify the most promising life-detection targets for the next decade.

Habitability is a trajectory, not a snapshot.

The Thread of the Course

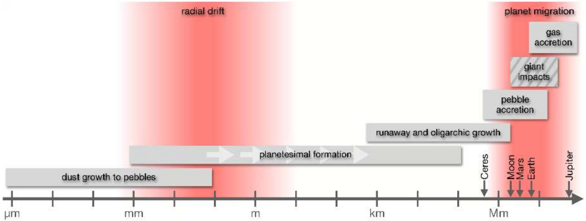
- L1–L2: solar system + protoplanetary disks.
- L3–L4: heat, energy, differentiation, magnetospheres.
- L5–L6: atmospheres I + II.
- L7–L8: surfaces + interiors.
- L9–L12: solar system body by body.
- L13: exoplanets.
- Today: **step back**. Same physics everywhere; what does the integrated picture look like?

1. Solar System in the Exoplanet Context



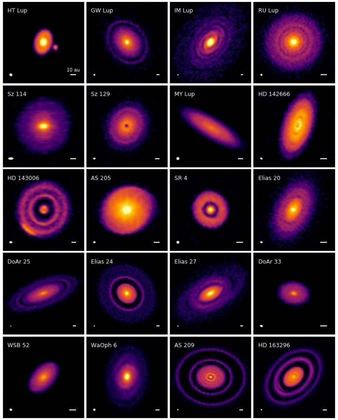
Dust-to-planet growth. Drazkowska et al. (2023)

Formation Theory Meets Observation

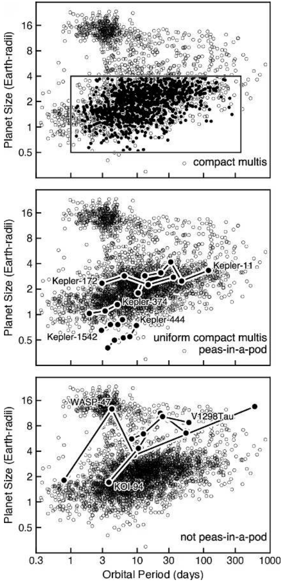


- Dust → pebbles → planetesimals → embryos.
- Streaming instability + pebble accretion + runaway envelope accretion.
- Disk lifetime 3–5 Myr is the deadline.
- ALMA disk substructure is direct observational input.

ALMA's DSHARP Reveals Forming-Planet Signatures



- 240 GHz continuum, 20 nearby disks.
- Concentric rings, gaps, asymmetries: nearly ubiquitous.
- Most are interpreted as signatures of in-progress planet growth.
- Andrews+ 2018.

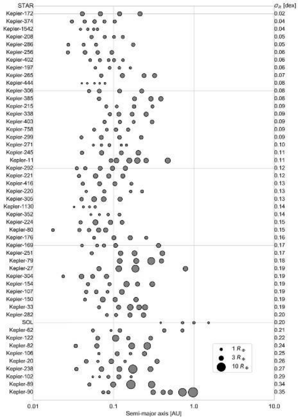


Period-radius distribution of confirmed exoplanets (top); compact-multi vs peas-in-a-pod vs not-peas-in-a-pod classification (bottom). Raymond & Morbidelli (2022).

Where the Solar System is Atypical

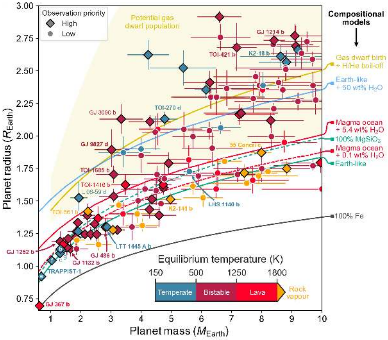
1. **No super-Earth or sub-Neptune.** The most common single class of planet, and we have none.
2. **Wide, low-eccentricity giant orbits.** Many exoplanet giants are eccentric, inclined.
3. **No hot Jupiter, no hot Neptune.** Both classes occur in ~ 0.5–1% of systems but not in ours.
4. **Irregular inner-system spacing.** Solar system terrestrials span 25× in mass; not a peas-in-a-pod.

Peas-in-a-Pod: Solar System Doesn't Fit



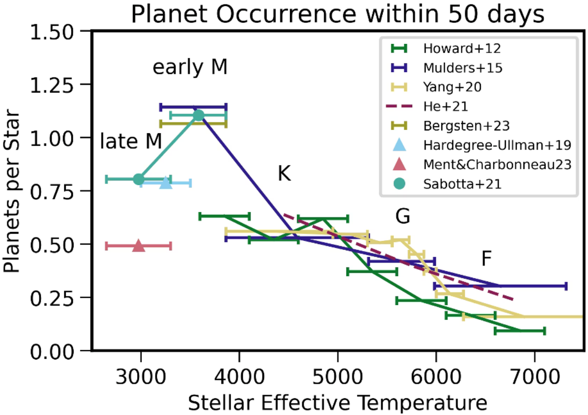
- Compact multi-systems with ≥ 4 transiting planets within 1.52 AU.
- Ranked by uniformity of planet sizes.
- Solar system: *bottom quintile* of size uniformity.
- Inner solar system did not produce the typical compact-multi architecture.

Mass-Radius: Where Each Planet Sits



- Lichtenberg 2025.
- EOS tracks: 100% Fe, Earth-like, $MgSiO_3$, water-rich, magma + H_2O, gas dwarf.
- Solar system terrestrials sit on the Earth-like curve.
- Many close-in super-Earths sit there too: *stripped cores*.
- Larger planets carry volatile envelopes that swell their radii.

Occurrence Climbs Toward M Dwarfs



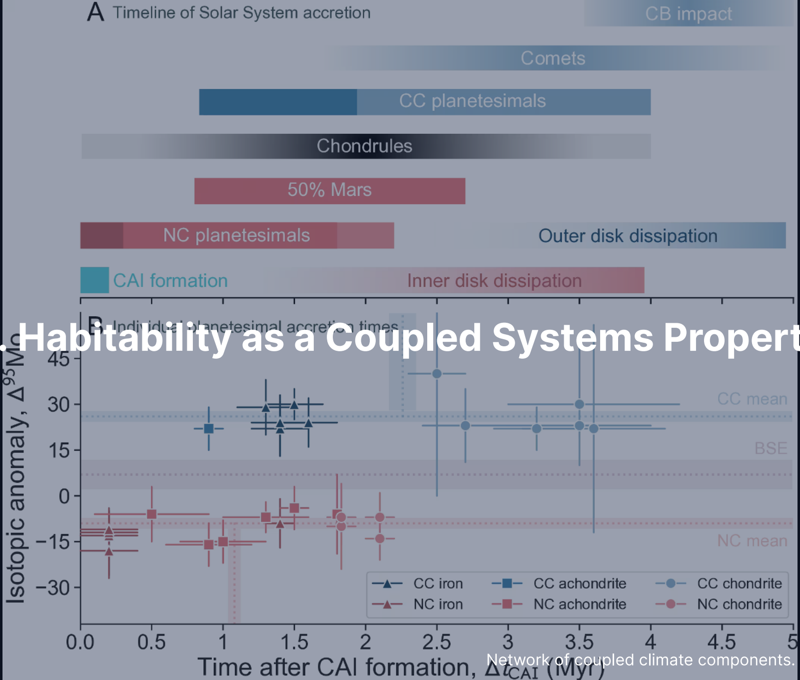
- Small planets (1–4 $R_\oplus$, P < 50 d) per star vs T_{eff}.
- Occurrence roughly 2× higher around early M dwarfs than F dwarfs.
- Trend continues into late M.
- Mulders+ 2024.
- **Earth-sized planets are common around the most numerous class of stars.**

Is the Solar System Rare?

Honest answer: **not obviously, but not yet known**.

- Long-period detection floor set by survey duration. Kepler had only 4 yr.
- Earth analogues at 1 AU around G dwarfs: at the edge of current sensitivity.
- PLATO (2026) + Gaia DR4 (late 2026) will resolve.
- Confident now: inner solar system is *atypical* relative to known compact systems.
- Open: whether this is bad luck, Jupiter's fingerprint, or the typical outcome around stars we haven't sampled yet.

2. Habitability as a Coupled Systems Property

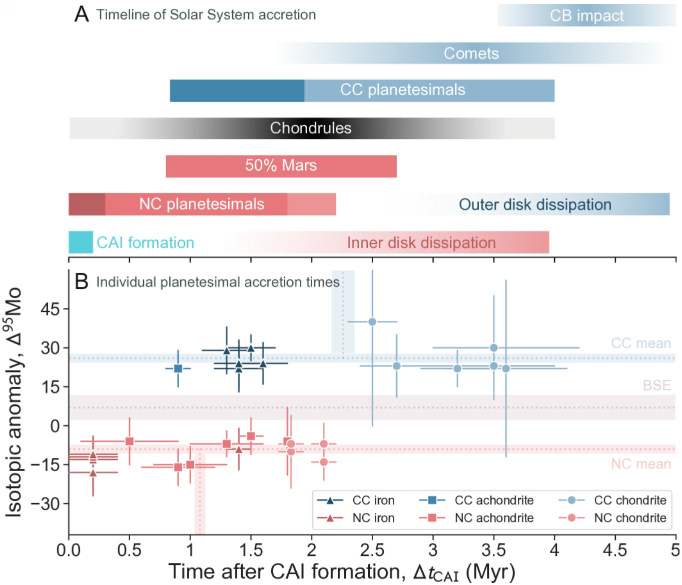


Network of coupled climate components. Lichtenberg et al. (2023)

The Stack of Requirements

Habitability is not a checklist. It is a stack of necessary couplings:

- **Star**: stable luminosity over Gyr; not too active.
- **Orbit**: in HZ; eccentricity not too high.
- **Planet bulk**: massive enough to retain atmosphere + drive interior.
- **Atmosphere**: temperature, pressure, escape rate compatible with surface water.
- **Interior**: heat budget drives outgassing + dynamo.
- **Surface + tectonics**: long-term volatile recycling.
- **Biosphere**: once present, modifies all of the above.

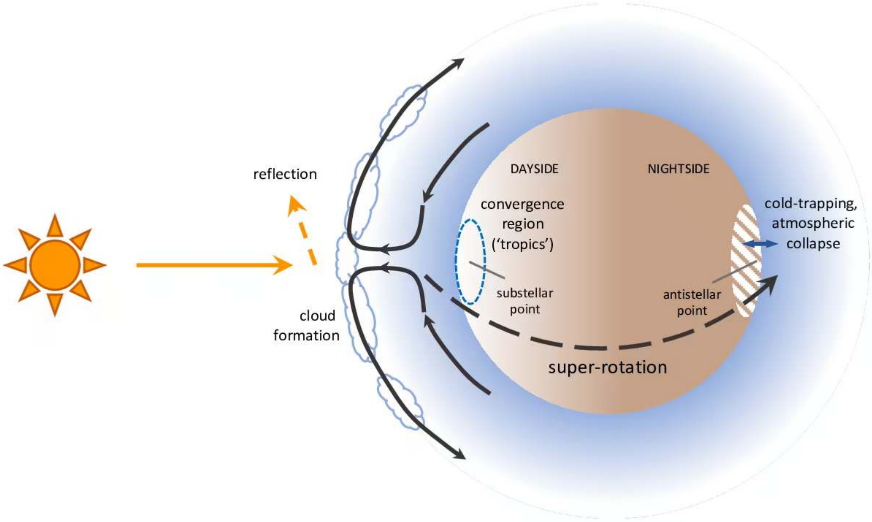


Network of coupled climate components: interior ↔ atmosphere ↔ surface ↔ biosphere over Gyr timescales. Lichtenberg et al. (2023).

Reading the Coupling Stack

- Interior ↔ atmosphere: outgassing, volatile partitioning, magma-ocean chemistry.
- Atmosphere ↔ surface: weathering, photochemistry, redox state of near-surface volatiles.
- Surface ↔ interior: tectonic recycling, sequestration of carbonates and water.
- Star ↔ atmosphere: irradiation, XUV-driven escape, photochemical loss.
- Biosphere (once present) feeds back on atmosphere, surface, and ocean chemistry.

Break any link, and the system can drift into a regime where surface liquid water is no longer thermodynamically possible.

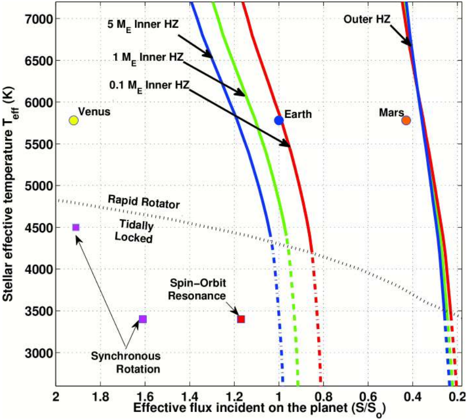


Atmospheric circulation on a tidally locked rocky planet: irradiation, cloud formation, super-rotation, dayside-nightside heat transport. Wordsworth & Kreidberg (2022).

Reading the Tidally Locked Circulation Diagram

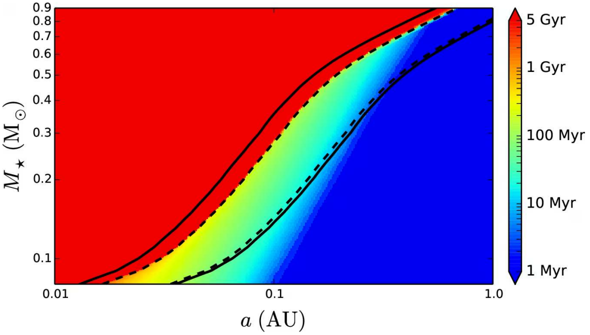
- Permanent dayside receives all stellar irradiation; partial reflection sets the planetary albedo.
- Substellar convergence region: convection lifts moisture, forms thick clouds.
- Equatorial super-rotation transports heat from dayside to nightside.
- Permanent nightside acts as a cold trap; condensable volatiles can freeze out.
- Atmospheric collapse possible when day-night transport is insufficient to keep volatiles vapour.

The HZ is Not a Line



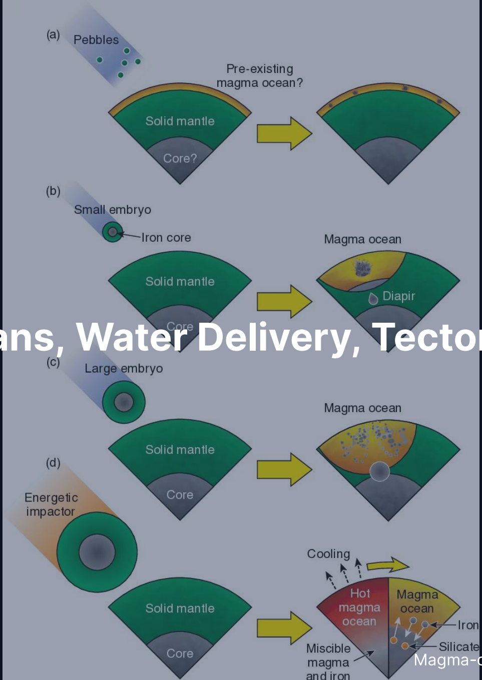
- Inner edge: runaway greenhouse.
- Outer edge: maximum CO_2 greenhouse.
- Two planets at the same flux can have radically different climates (Earth vs Venus).
- Time-dependent: stars brighten over MS lifetime.
- HZ planet today ≠ habitable trajectory.

M Dwarf HZ: Pre-MS Runaway Phase



- Duration of runaway greenhouse on water-bearing planet vs $M_\star$ + a.
- Late M dwarfs: *Gyr* of runaway before star reaches MS.
- Habitable-zone planet started *inside* runaway zone.
- Loses much of initial water before life could plausibly start.
- Luger & Barnes (2015).

3. Magma Oceans, Water Delivery, Tectonic Thermostat

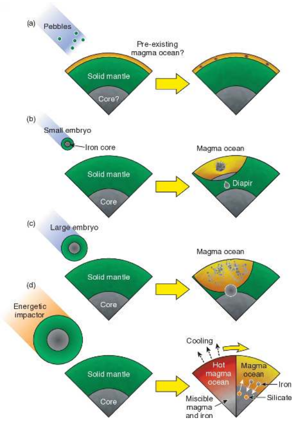


Magma-ocean differentiation. Lichtenberg et al. (2023)

Why ”Delivery” Isn’t the Right Frame

- Earth, Venus, Mars likely received *similar* initial volatile inventories.
- Yet present-day water inventories differ by orders of magnitude.
- Difference: **what each body did with its initial inventory**.
- Four levers (in order of importance):
  - Magma ocean partitioning (core vs mantle vs steam atmosphere)
  - Magma ocean atmospheric escape
  - Mantle-atmosphere exchange over Gyr
  - Stellar evolution history

Magma Ocean Differentiation Sets the Boundary Conditions

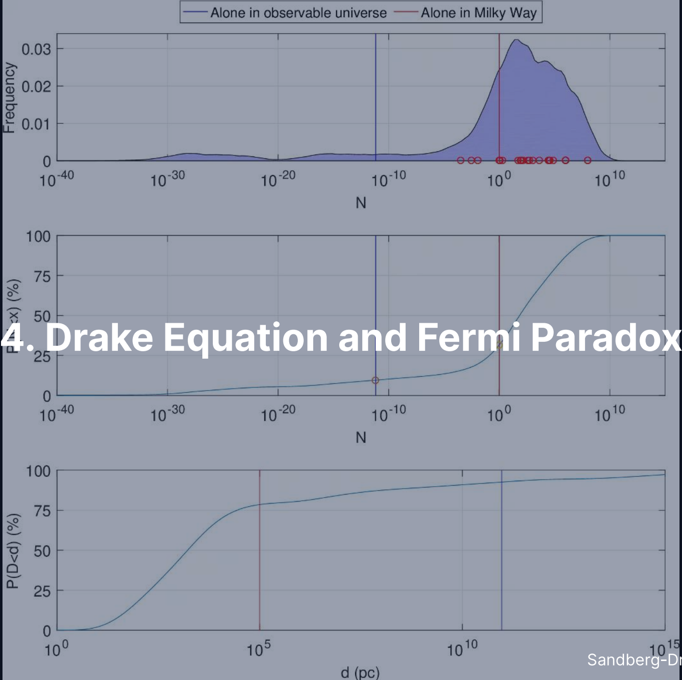


- Initially molten silicate mantle separates from denser metal.
- Volatiles partition between mantle and steam atmosphere.
- Fraction sequestered vs outgassed depends on redox state, solidification timescale, depth.
- *Identical* starting inventories → very different end states.
- Lichtenberg et al. (2023).

Tectonic Regime: The Long-Term Thermostat

- Volcanism injects CO_2; rainfall + silicate weathering removes it; subduction returns carbonates to mantle.
- Negative feedback: warmer climate → faster weathering → cooler subsequent climate.
- Three conditions, all required:
  - Liquid water
  - Active volcanism
  - Tectonic recycling of carbonates back to the mantle
- Venus failure mode: surface dries → no weathering sink → CO_2 accumulates to 92 bar.
- For exoplanets: tectonic regime is *unmeasurable remotely* with current/near-term tools.

4. Drake Equation and Fermi Paradox



Sandberg-Drexler-Ord Drake posterior (2018)

The Drake Equation

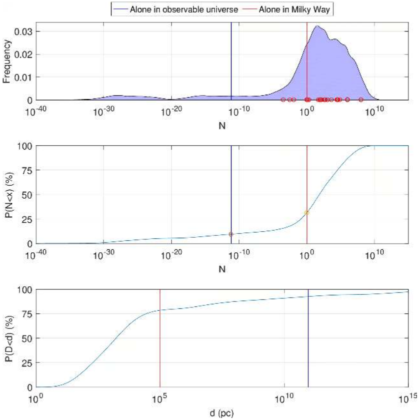
$$N = R_\star \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

- Only the first three factors are now measured.
  - $R_\star$: stellar formation rate. Astrophysics.
  - f_p: fraction with planets. Of order unity (Kepler).
  - n_e: HZ planets per system. ~ 0.1–0.6 for Sun-like (Bryson+ 2021).
- f_l, f_i, f_c, L: *not* constrained by orders of magnitude.
- Drake equation = framing tool, **not an estimator**.

Five Reasons Not to Quote a Number for *N*

1. Factors are not independent. Biospheres co-evolve with atmospheres.
2. Most factors span 4–10 orders of magnitude in published estimates.
3. Treats a contingent process as a steady-state pipeline.
4. Implicitly assumes uniformity over diverse planets and stars.
5. Anthropic selection in the only data point we have.

Sandberg+ 2018: Honest Posterior Dissolves Fermi



- Monte Carlo each Drake factor over its full published range.
- Result is *bimodal*.
- ~ 1/3 chance of being alone in the Milky Way (CDF at N < 1).
- ~ 10% chance of being alone in the entire observable universe (CDF at $N \ll 1$).
- Apparent silence is unsurprising once priors are honest.

Why Bimodal? Log-Sum of Broad Priors

Take the log:

$$\log_{10} N = \log_{10}(R_\star f_p n_e) + \log_{10} f_l + \log_{10} f_i + \log_{10} f_c + \log_{10} L$$

- Each soft factor has log-uniform prior over many decades.
- Sum of log-uniform variables = quasi-Gaussian by central limit theorem.
- Width: ~ 30 decades. Tail mass below $N = 1$ is large.
- Lesson: **broad priors on multiplicative factors generate broad and skewed product distributions.**

Fermi Paradox: A Question, Not an Answer

Possible resolutions:

- **Rare-life**: f_l very small.
- **Rare-intelligence**: f_i is the bottleneck.
- **Great filter ahead**: L short; civilisations self-terminate.
- **Detection threshold**: signals exist, are missed.
- **Zoo / quiet**: civilisations conceal themselves or use signalling we cannot recognise.

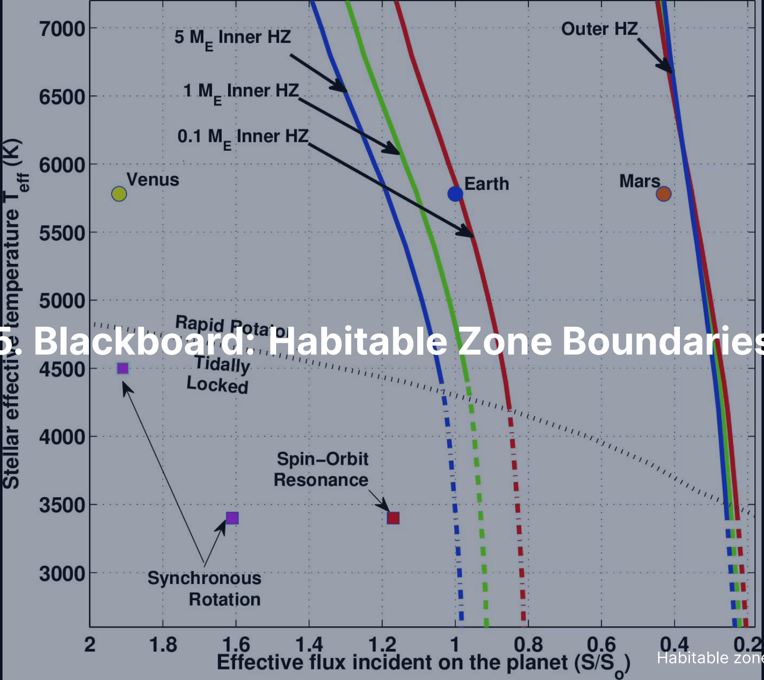
Use Fermi paradox as a check on intuition about Drake-equation factors. *Not* as evidence.

Break

End of Part 1: Solar System Context + Habitability

Part 2: Habitable Zone Derivation, Astrobiology, Life Detection

5. Blackboard: Habitable Zone Boundaries



Habitable zone vs T_{eff}. Kopparapu (2014)

Setup: Stellar Flux and Equilibrium Temperature

Star with bolometric luminosity $L_\star$, planet at distance d:

$$F_\star(d) = \frac{L_\star}{4\pi d^2}$$

Bond albedo A_B, fast-rotating sphere, equilibrium:

$$T_{\mathrm{eq}}(d) = \left[\frac{L_\star(1 - A_B)}{16\pi\sigma\epsilon\, d^2}\right]^{1/4}$$

For Earth: $A_B = 0.3$, $\epsilon = 1$, $L_\star = L_\odot$, $d = 1$ AU $\rightarrow T_{\mathrm{eq}} \approx 255$ K.

Surface is 288 K because of ~ 33 K of greenhouse warming.

Effective Stellar Flux

Define S_{eff} = TOA flux at HZ boundary / $S_{\oplus}$ = 1361 W m^{-2}.

$$d = (1\,\mathrm{AU})\sqrt{\frac{L_\star / L_\odot}{S_{\mathrm{eff}}}}$$

Read S_{eff} off Kopparapu+ 2013 1D radiative-convective calculation:

- Inner edge (Sun-like): $S_{\mathrm{in,eff}} \approx 1.06$.
- Outer edge (Sun-like): $S_{\mathrm{out,eff}} \approx 0.35$.

Inner Edge: Simpson-Nakajima Limit

In a moist atmosphere, OLR is bounded above:

$$F_{\mathrm{OLR}}^{\max} \approx 280\ \mathrm{W\,m^{-2}}$$

- Once absorbed flux > this, no radiative equilibrium possible.
- Oceans driven into the atmosphere → runaway greenhouse.
- Sun-like inner edge:

$$d_{\mathrm{in}} = (1\ \mathrm{AU})/\sqrt{1.06} \approx 0.97\ \mathrm{AU}$$

Venus at 0.72 AU receives $S_{\mathrm{eff}} \approx 1.9$ — *well inside* the inner edge.

Outer Edge: Maximum CO_2 Greenhouse

At low T + high $p_{\mathrm{CO_2}}$, CO_2 condenses out as ice clouds.

- Maximum greenhouse warming saturates.
- No matter how much CO_2 you add, surface stays cold.
- Sun-like outer edge:

$$d_{\mathrm{out}} = (1\ \mathrm{AU})/\sqrt{0.35} \approx 1.69\ \mathrm{AU}$$

Mars at 1.52 AU is just inside the outer edge.

HZ Across Stellar Types

HZ width $\propto \sqrt{L_\star}$.

- **G dwarf** ($L_\odot$): d = 0.97–1.69 AU. Lifetime ~ 10 Gyr.
- **K dwarf** (0.3 $L_\odot$): d = 0.53–0.93 AU. Lifetime 24 Gyr. *May be most habitable class* (Cuntz & Guinan 2016).
- **M dwarf** ($5 \times 10^{-4}\,L_\odot$, e.g. TRAPPIST-1): d = 0.022–0.038 AU. Inside Mercury’s orbit. Lifetime 10^{12} yr but: tidally locked + Gyr pre-MS runaway.

Main-Sequence Lifetime: A One-Line Scaling

$E_{\rm nuc} \propto M_\star$, $L_\star \propto M_\star^{3.5}$:

$$t_{\rm MS} \sim \frac{E_{\rm nuc}}{L_\star} \propto M_\star^{-2.5}$$

Normalising on Sun ($t_{\rm MS,\odot} \approx 10$ Gyr):

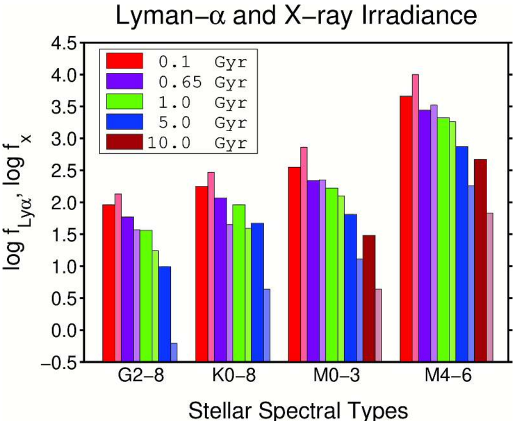
$$t_{\rm MS} \approx 10\,(M_\star/M_\odot)^{-2.5}\ \mathrm{Gyr}$$

- $0.7\,M_\odot$ K dwarf: ≈ 24 Gyr.
- $0.5\,M_\odot$ early M: ≈ 57 Gyr.
- $0.1\,M_\odot$ late M: ≈ 3000 Gyr.

Cooler stars give biology more time.

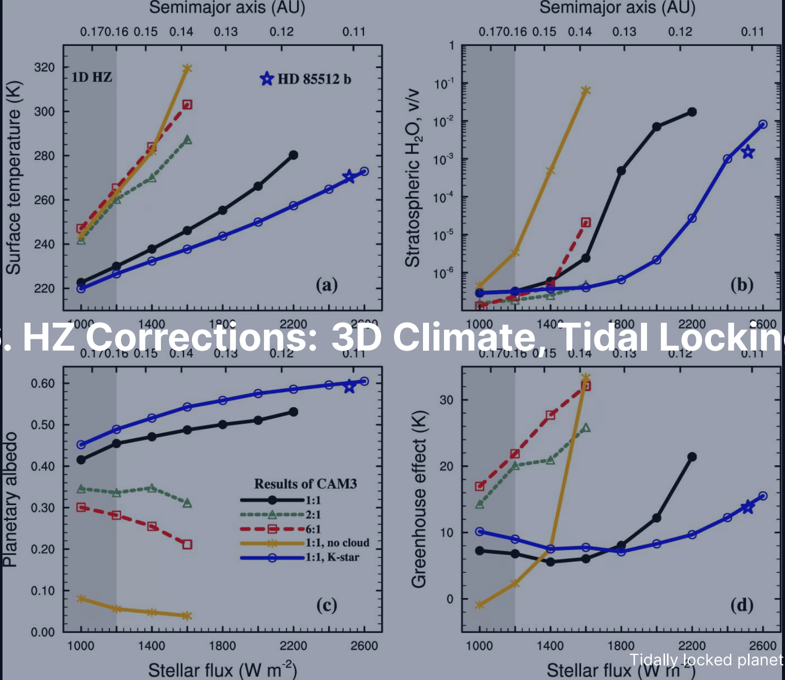
$L \propto M^{3.5}$ is a single-power-law approximation; piecewise relations give $\sim M^4$ for $M \gtrsim 0.4\,M_\odot$ and $\sim M^{2.3}$ below. Numbers good to factors of a few.

XUV at the HZ: M Dwarfs are Hostile



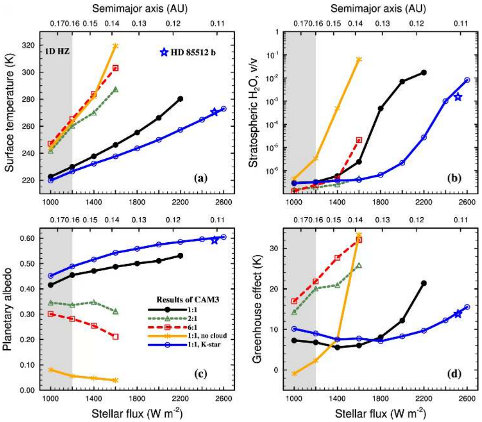
- Lyman-α + X-ray at HZ orbit, vs spectral type and age.
- M dwarfs deliver 10^2–$10^3\times$ more EUV/X-ray to HZ planets than G dwarfs.
- High-flux phase lasts *several Gyr*.
- K dwarfs: elevated for ~ 100s Myr, then declining; less harsh than M, harsher than G.
- Cuntz & Guinan (2016).

6. HZ Corrections: 3D Climate, Tidal Locking



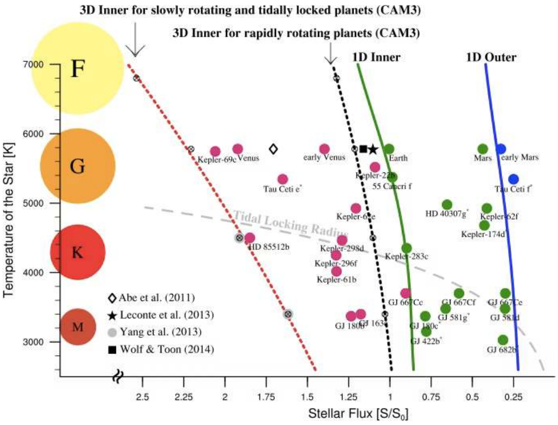
Tidally locked planet clouds. Yang et al. (2013)

3D GCMs Shift the Inner Edge of the M Dwarf HZ



- Yang+ 2013: tidally locked planet GCM with clouds.
- Substellar convection lifts thick water clouds.
- Albedo rises to ~ 0.6.
- Stabilising feedback against runaway.
- Inner edge tolerates ~ 2× higher stellar flux.
- 1D HZ underestimates M dwarf habitable region.

Where M Dwarf HZ Planets Actually Sit



- Shields+ 2016 review.
- Solid lines: 1D Kopparapu boundaries.
- Dashed/dotted: 3D GCM corrections.
- Grey curve: tidal-locking radius.
- M dwarf HZ planets all sit *on or inside* the locking radius.
- → tidally locked is the rule.

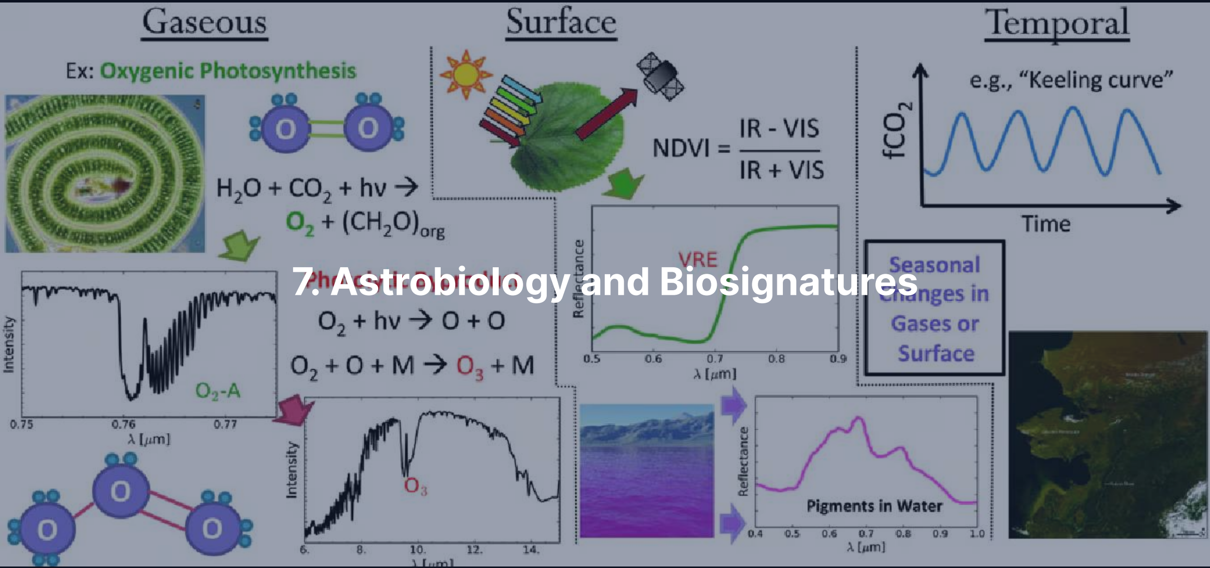
Tidal-Locking Timescale

$$\tau_{\mathrm{lock}} \sim \frac{\omega_0 \, a^6 \, I_p \, Q_p}{3 \, G \, M_\star^2 \, k_{2,p} \, R_p^5}$$

Strong a^6 dependence:

- Earth at 1 AU: $\tau_{\mathrm{lock}} \sim 10^{12}$ yr (much longer than universe age).
- TRAPPIST-1 b ($a \approx 0.011$ AU): $\sim 10^7$ yr or shorter.
- TRAPPIST-1 system age: ~ 8 Gyr.
- Every M dwarf HZ planet has had time to lock.

7. Astrobiology and Biosignatures



Biosignature classes. Schwieterman et al. (2018)

What is Life?

Working definition (Joyce 1994, building on Lederberg 1965 + Viking-era debate):

- A self-replicating, metabolising, evolving chemical system.
- Operational, not philosophically deepest.
- Lets us decide what to look for.

Chemical baseline:

- Carbon-based, water-as-solvent.
- Only example we have.
- Dominates the search because we know its spectroscopic + geochemical fingerprints.

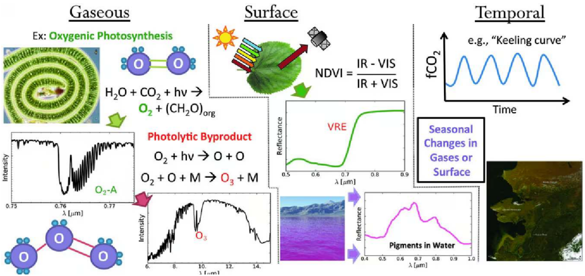
Origin of Life: Four Live Hypotheses

Not solved. Present as competing, not as a settled story.

- **RNA world**: ribozyme-driven, RNA precedes proteins. Hard prebiotic chemistry of RNA.
- **Metabolism-first / alkaline hydrothermal vents**: autocatalytic cycles precede information storage. Geochemistry compatible.
- **Surface "warm little ponds"**: wet-dry cycling concentrates and polymerises. Darwin 1871; revived by experiment.
- **Panspermia**: relocates the problem, doesn't solve it.

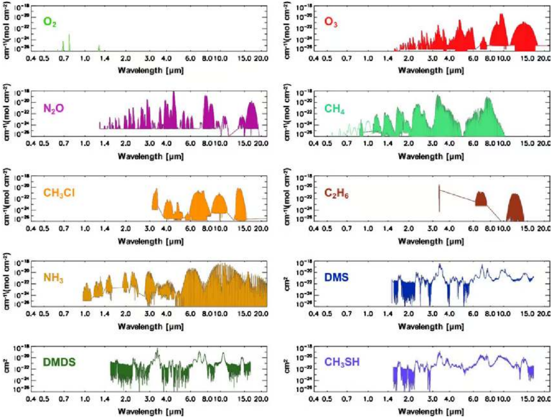
Earliest direct evidence on Earth: ~ 3.5 Ga. Possible to ~ 4.0 Ga (contested).

Three Classes of Biosignature



- **Gaseous**: O_2, O_3, CH_4, N_2O, organosulphur, organohalogen.
- **Surface**: vegetation red edge near 700 nm.
- **Temporal**: seasonal cycles, e.g. Earth's CO_2 Keeling curve.
- Convincing claim should combine ≥ 2 classes.
- Schwieterman+ (2018).

Biosignature Gases in the Mid-IR



- Ten potential biosignature gases.
- Each has a characteristic IR absorption signature.
- O_3: 9.6 μm. CH_4: 3.3 + 7.7 μm. N_2O: across mid-IR.
- DMS / DMDS: 3–15 μm window.
- Drives the LIFE / HWO wavelength designs.

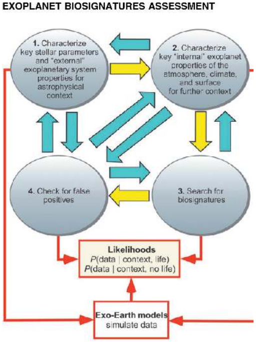
Disequilibrium is the Diagnostic

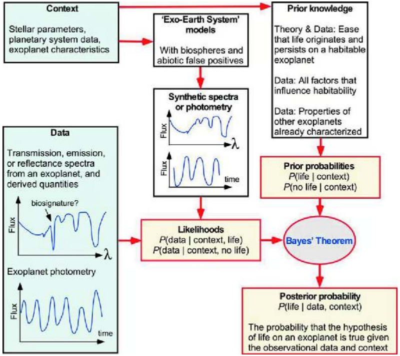
Single gas in isolation: rarely convincing.

Most diagnostic: *disequilibrium combinations*.

- Earth: O_2 (21%) + CH_4 (1.8 ppm). React on ~decade timescales.
- Joint presence requires continuous biological resupply (oxygenic photosynthesis + methanogenesis).
- Pre-Great Oxidation Event Earth: CH_4 + N_2O or CH_4 + CO_2.
- Krissansen-Totton+ 2018 formalised disequilibrium approach.

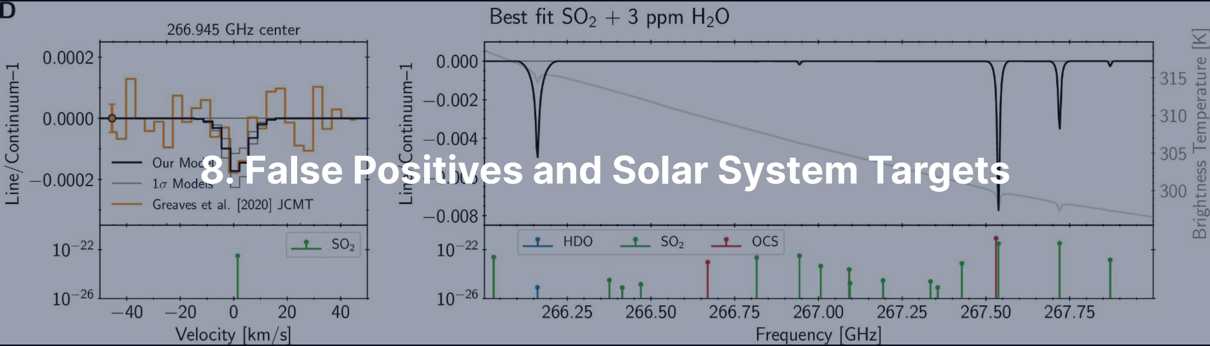
Catling 2018: A Bayesian Framework





Detection is a quantitative inference from a chain of measurements: priors on host star, planet, atmosphere, false-positive pathways. Not a yes/no statement on a single observation.

8. False Positives and Solar System Targets



Venus phosphine controversy. Lincowski et al. (2021)

False Positives: Every Gas Has an Abiotic Pathway

- **Abiotic O_2**: H_2O photolysis + H escape (Wordsworth+ 2014, Luger & Barnes 2015). CO_2 photolysis.
- **Abiotic CH_4**: serpentinisation of olivine + water. Volcanic outgassing at low $f_{\mathrm{O_2}}$.
- **Abiotic N_2O**: lightning chemistry, photochemistry.
- **Abiotic DMS**: no significant Earth source known. But ”not on Earth” ≠ ”impossible elsewhere”.

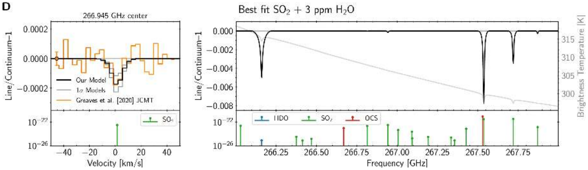
Biosignature detection is an inverse problem: must rule out *all* plausible abiotic pathways.

Solar System Targets for Life Detection

Five solar-system bodies under active astrobiological assessment:

- **Mars**: past habitability established. Mars Sample Return is the path forward.
- **Europa**: confirmed subsurface ocean. Clipper at Jupiter 2030.
- **Enceladus**: plumes deliver ocean chemistry (H_2 + organics + phosphates) to your spacecraft. Decadal-Survey priority.
- **Titan**: complete hydrocarbon hydrology + transient impact-melt water. Dragonfly arrives 2034.
- **Venus cloud layer**: phosphine claim contested.

Venus Phosphine: A Cautionary Case Study

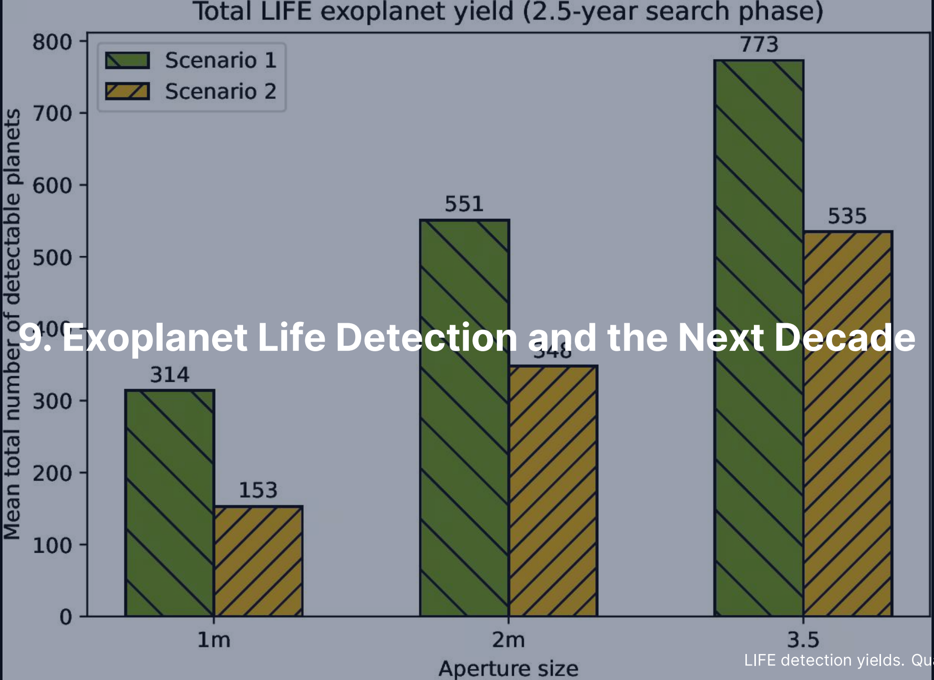


- Greaves+ 2021: PH_3 in Venus clouds claimed as biosignature.
- Lincowski+ 2021: same data fits SO_2 in Venus mesosphere.
- Original feature is at the JCMT/ALMA noise floor.
- Direct analogue of the K2-18 b DMS controversy.
- DAVINCI + EnVision + VERITAS will resolve.

Venus Phosphine + K2-18 b: Same Lesson, Two Telescopes

- Both: data at edge of instrumental sensitivity.
- Both: molecular identification contested.
- Both: plausible abiotic explanations not ruled out.
- Right response in either case: *better data*, ideally from independent instruments.
- Both episodes have driven the design of next-generation instruments.
- Pedagogical value is high regardless of how each turns out.

9. Exoplanet Life Detection and the Next Decade

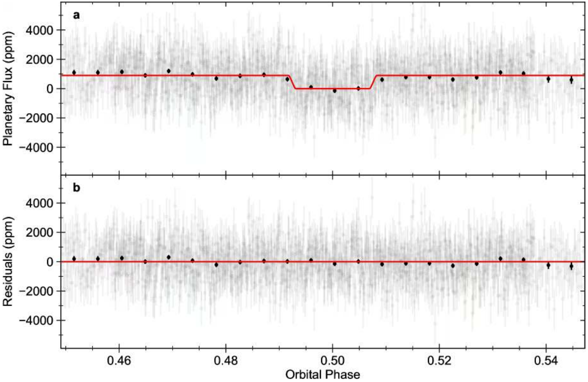


LIFE detection yields. Quanz et al. (2022)

What Convincing Exoplanet Biosignature Detection Requires

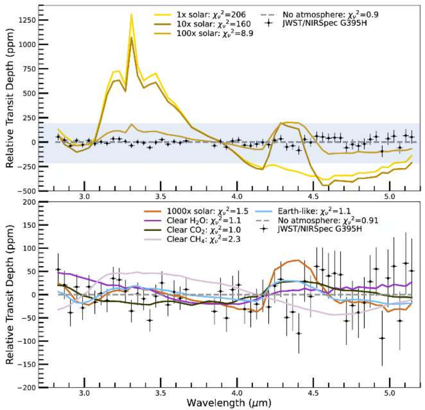
1. Multiple independent gases in disequilibrium (oxidising + reducing).
2. Temporal variability consistent with a biological cycle.
3. Geological context (mass, radius, host, age, escape) excluding obvious abiotic pathways.
4. Independent confirmation by different instrument / wavelength / observatory.

TRAPPIST-1 b: What JWST Can Already Tell Us



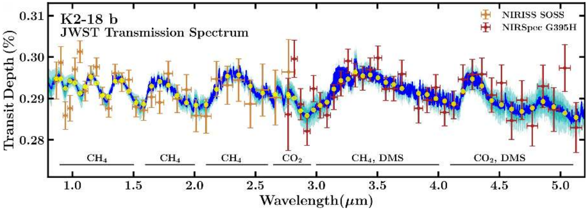
- MIRI 15 μm secondary eclipse.
- $T_d \approx 508$ K = bare-rock equilibrium.
- No thick atmosphere to redistribute heat.
- Greene+ 2023.
- M dwarf HZ planets often retain little atmosphere.

LHS 475 b: Featureless Spectrum



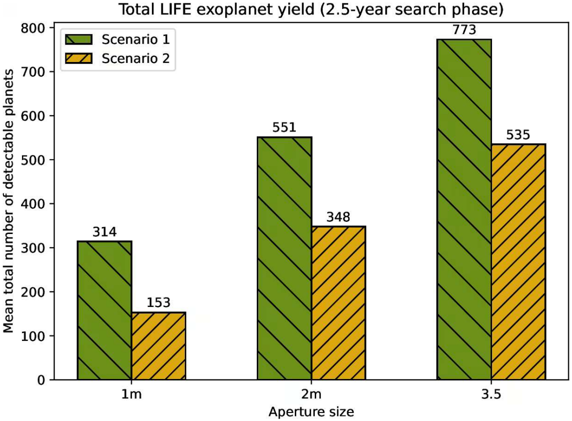
- NIRSpec G395H, Earth-sized M dwarf planet.
- Featureless transmission.
- H_2-dominated ruled out.
- High-mean-molecular-weight or airless: still consistent.
- Lustig-Yaeger+ 2023.

K2-18 b: How Tentative Biosignatures Are Tested



- Madhusudhan+ 2023: CH_4 + CO_2 + tentative DMS at ~ 3σ.
- Hycean (H_2 over global ocean) interpretation contested.
- Multiple groups failed to reproduce DMS.
- Whatever the resolution: **textbook case in real-time biosignature testing**.

LIFE: Mid-IR Nulling Interferometer



- Four free-flying spacecraft, 4–18 μm.
- Nulling: stellar light suppressed by destructive interference.
- 3.5 m aperture: 500–800 planets, ~ 25 rocky HZ.
- Quanz+ 2022.
- Companion to HWO (visible/NIR).

The 2026–2040 Mission Queue

- **Now–2030**: JWST continues. BepiColombo at Mercury 2026. PLATO 2026. Roman 2027. Dragonfly launch 2028. Ariel 2029. Europa Clipper at Jupiter 2030.
- **2030s**: JUICE Ganymede orbit. Dragonfly at Titan 2034. Ariel atmospheric statistics. ELT class first light 2028+. DAVINCI/EnVision/VERITAS at Venus. Mars Sample Return delivery (re-baselined). Uranus orbiter concept advances.
- **2040s**: HWO + LIFE in hardware. First plausible direct atmospheric characterisation of *Earth-analogue* exoplanets.

The first time ”are we alone?” is empirical, not philosophical. The empirical regime begins within the careers of the students in this room.

The Five Biggest Things We Have Learned

1. Planet formation is a **physical process**, not chance. Same laws everywhere.
2. Planetary interiors are **heat engines** driving tectonics, dynamos, atmospheres on Gyr timescales.
3. Atmospheres are **not static**. They form, evolve, escape, and feed back on surfaces and biospheres.
4. Habitability is a **coupled systems property**, not a single condition.
5. The solar system is **one example** in a much larger statistical population.

The Five Biggest Open Questions

1. How does life originate, and how often?
2. What sets the radius valley?
3. When did Jupiter form, and was it the architect of the inner solar system?
4. Was Mars ever inhabited?
5. Is the solar system rare or typical?

Final Framing

- Planetary science is now *the science of comparative climate, interior, and life-hosting trajectories*.
- Solar system: reference, not benchmark.
- Exoplanet population: statistical context.
- The frontier is moving fast.
- One systems-level view: habitability, formation, escape, weathering, biology are *one coupled problem*, with one consistent set of physical principles, the same set wherever you go in the galaxy.

That is what the course has been about.

Thank You

Introduction to Planetary Science is complete.

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`